

CHAPTER 2

SURVEY OF TECHNOLOGIES

2.1 Frontend:

The frontend for **MealMentor AI** is developed as an intelligent Android mobile application responsible for capturing food photographs, uploading medical lab reports, rendering real-time interactive nutritional charts, and hosting the conversational AI Dietitian interface.

2.1.1 Comparison with Other Frameworks

Table 2.1: Comparison of Frontend Mobile Frameworks

ASPECT	REACT NATIVE (EXPO)	FLUTTER	NATIVE ANDROID (KOTLIN / JAVA)
Development Speed	Faster due to JavaScript/JSX ecosystem, instant Hot Reloading, and pre-built Expo modules.	Good, but requires learning Dart and complex widget tree hierarchies.	Slower, requiring separate codebases for Android and iOS.
Performance	Near-native performance as it compiles to native views with direct JSI bridge execution.	Near-native performance compiled directly to machine code via Skia engine.	Excellent, as it is platform-native direct hardware code.
User Interface (UI)	Consistent, highly customizable UI using Flexbox, organic squircles, and glassmorphic blurs.	Consistent UI on all platforms thanks to its own rendering engine.	Excellent UI customization, but requires duplication of effort per platform.
Codebase	Single codebase for Android, iOS, and Web.	Single codebase for iOS, Android, and Desktop.	Two separate codebases (Kotlin for Android, Swift for iOS).

2.1.2 React Native & Expo

2.1.2.1 What is React Native?

React Native is an open-source mobile application framework created by Meta for building natively compiled mobile applications from a single JavaScript/TypeScript codebase.

- **Expo SDK:** Managed toolset providing streamlined camera access, image picking, and file management without complex native build configurations.
- **React Navigation:** Manages seamless stack and bottom-tab screen routing across dashboards and chat views.
- **React Native Chart Kit:** Renders responsive SVG charts (line graphs, donut macro rings, and radar charts).
- **React Context & Axios:** Handles global application state and asynchronous REST API communication.

2.1.3 Features of the Frontend Stack

- **Cross-Platform Compatibility:** Single codebase targeting Android with direct extensibility to iOS.
- **Rich UI Components:** Organic squirele cards, glassmorphic frosted overlays, and vibrant progress rings designed for health and diet tracking.
- **Hot Reload & Fast Refresh:** Instantaneous code reflection drastically speeding up development and UI fine-tuning.
- **Real-Time Visualization:** Dynamic recalculation of daily calorie budgets and macro bars upon logging meals.

2.1.4 Advantages

- **Cost-Effective & Rapid Development:** Single codebase eliminates duplicate engineering efforts.
- **High Performance:** Fluid 60 FPS animations and fast screen transitions.
- **Extensive Ecosystem:** Access to thousands of battle-tested NPM libraries for camera, charts, and storage.

2.1.5 Disadvantages

- **Bridge Overhead:** High-frequency image manipulations require asynchronous background transfers.
- **Native Modules:** Deep platform-specific hardware customizations require custom native configurations.

2.2 Backend:

The backend for MealMentor AI is developed using **Node.js** and the **Express.js** framework to create a high-throughput, non-blocking asynchronous RESTful API server.

- **Node.js:** A robust, event-driven JavaScript runtime built on Chrome's V8 engine.
- **Express.js:** A minimalist web framework providing flexible routing and middleware chains.
- **JSON Web Tokens (JWT) & Bcrypt:** Provides stateless user authentication and salted cryptographic password hashing.
- **Multer:** Handles multipart stream uploads for food photographs and medical lab reports.

2.2.1 Advantages of the Backend Stack

- **High Concurrency:** Non-blocking asynchronous I/O efficiently handles simultaneous upload and AI inference requests.
- **Unified Language:** Single language (JavaScript/TypeScript) shared across mobile client and server.
- **Lightweight Footprint:** Low memory consumption enabling reliable hosting on cost-effective student and cloud tiers.

2.2.2 Disadvantages

- **Single-Threaded Execution:** Heavy local compute operations can block the event loop (mitigated by offloading AI processing to cloud AI APIs).

2.2.3 Database: PostgreSQL & pgvector

2.2.3.1 What is Relational Database with pgvector?

PostgreSQL is an advanced, enterprise-grade open-source relational database providing full ACID compliance. The **pgvector** extension enhances PostgreSQL by enabling native high-dimensional vector similarity indexing directly alongside traditional relational tables.

Table 2.2: Comparison of Database Management Systems

ASPECT	POSTGRESQL + PGVECTOR	MONGODB (NOSQL)	MYSQL (RELATIONAL)
Data Integrity	Strict ACID compliance with relational constraints.	Document-based schema-less consistency.	Strict ACID compliance.
Vector AI Search	Native built-in vector similarity search.	Requires paid Atlas Vector Search add-on.	Requires external vector database.
Analytics & Joins	Superior SQL joins and time-series aggregations.	Aggregation pipeline, less optimal for multi-joins.	Standard SQL joins, no vector indexing.

2.2.3.2 Features & Advantages

- **Unified Relational & Vector Storage:** Manages users, meal logs, and AI profile embeddings within a single database, eliminating the need for a separate vector database.
- **Array & JSON Support:** Native array types store dietary preferences and health tags efficiently.

2.3 Multimodal Artificial Intelligence & Cloud Services

The core intelligence of MealMentor AI is driven by modern multimodal foundation models. For academic projects, **Google Gemini Flash** (via Google AI Studio Free Tier) and **OpenAI GPT-mini** provide state-of-the-art multimodal vision with zero upfront cost and low latency.

Table 2.3: Comparison of Multimodal AI Models

ASPECT	GOOGLE GEMINI FLASH	OPENAI GPT-MINI	LEGACY MODELS (GPT-4)
Cost for Students	100% Free Tier (15 RPM / 1,500 RPD).	Ultra-low per-token cost.	High cost; costly for students.
Vision & Food Scan	Native multimodal vision for Indian dishes.	Solid vision understanding.	Good vision, high latency.
Document OCR	Extracts medical PDF lab tables accurately.	Good JSON extraction.	High cost per PDF page.
Inference Speed	Blazing Fast (<1.2s response time).	Fast (<1.8s response time).	Slower response times.

2.3.1 Multimodal Capabilities

- **Food Image Recognition:** Analyzes camera captures of Indian thalis (Paneer Butter Masala, Rotis, Dal Tadka) to extract calories, portion estimates (grams/katoris), and macronutrients (Protein, Carbs, Fats).
- **Lab Report OCR Parsing:** Ingests blood test PDFs and extracts standardized biomarkers (Fasting Sugar, HbA1c, Cholesterol, Vitamin D3).
- **Context-Aware AI Dietitian:** Provides personalized conversational dietary advice dynamically aligned with user lab reports and remaining calorie budgets.
- **Semantic Vector Embeddings:** Generates dense vector embeddings for RAG matching and personalized food recommendations.

2.4 Summarization

Table 2.4 presents the comprehensive summary of all surveyed technologies adopted across the MealMentor AI architecture.

Table 2.4: Master Technology Summarization

TECHNOLOGY	PURPOSE	FEATURES	ADVANTAGES	DISADVANTAGES
React Native & Expo	Frontend Mobile UI	Cross-platform, Hot Reload, SVG charting, native camera APIs	Fast development, native performance, unified UI	Bridge overhead for heavy local computations
Node.js & Express.js	Backend RESTful API	Event-driven, async I/O, modular middleware, JWT authentication	High concurrency, shared syntax, rapid development	Single-threaded CPU execution limit
PostgreSQL + pgvector	Database & Vector Engine	ACID compliance, relational schema, vector similarity search	Unified metadata & vector storage, high reliability	Requires basic vector index configuration
Google Gemini Flash	Multimodal AI & OCR	Food vision recognition, PDF lab report parsing, dietary chat	100% Free Tier, sub-second latency, native multimodal	Rate limits on free tier (15 RPM)
Cloud App Service & Storage	Hosting & Storage	Linux container hosting, encrypted object storage for assets	High availability, secure media persistence	Requires cloud account configuration

2.4.1 Summary of System Requirements

- **Client-Side:** Android OS device, Dual-Core processor, 3 GB RAM, Camera with autofocus, Internet connectivity.
- **Server-Side:** Node.js runtime, PostgreSQL with pgvector extension, Google AI Studio API Key, Cloud Hosting & Storage.